Explanations Using Defeasible Rulesets

Antony Baker**
BKRANT001@myuct.ac.za
University of Cape Town
Cape Town, Western Cape, South Africa

Abstract

Defeasible Rulesets are rulesets that embody a possible set of exceptions that when fired, trigger a specific label or prediction. The rulesets govern which exception chain will result in a prediction. A Defeasible Ruleset would be one in which a given set of implications typically results in another set of given conclusions. To that end, we want to explore and argue that exception chain predictions can provide a real foundation to build explanations that are built on a logical structure that can be clearly understood by humans. A well understood approach to Defeasible Reasoning is the framework defined by Kraus, Lehmann, and Magidor (KLM). The KLM framework has been used in the field of Knowledge Representation and Reasoning (KRR) to describe several inference algorithms. This paper explores the use of System W as an inference algorithm to compare against Exception Closed Conjunction Rule Sets (ECCRS). This paper evaluates the alignment of the ECCRS framework and the corresponding Most Specific Wins (MSW) predictions with the Defeasible closure algorithm. The paper aims to translate the ECCRS Most Specific Wins predictions into KLM style Knowledge Bases and compare the entailed predictions. The final explanation traces show that if a dataset satisfies 3 assumptions (sanity, no total override and strict global closure) then MSW prediction agrees exactly with System W inference as an inductive inference relation. Thus, we demonstrate why the ECCRS framework and its MSW predictions is validated by System W.

Keywords

Knowledge Representation and Reasoning, Artificial Intelligence, Symbolic classifiers, Defeasible Reasoning,

1 Introduction

Machine learning models are being increasingly deployed in consequential settings such as medical diagnosis, credit assessment, and criminal risk profiling. In these areas, merely providing a prediction alone is not sufficient. Users, regulators, and affected individuals need to understand not just what a model decided, but why. Post-hoc explanation methods, such as surrogate models, attempt to address this by approximating a black-box model with a simpler, interpretable one. However, as Rudin [12] argues, a surrogate is not the model it describes, and in high-stakes environments this gap is unacceptable. Symbolic rule-based classifiers offer an alternative. By representing decisions as explicit if-then rules over meaningful features, they make every prediction traceable by design, without requiring any additional explanation machinery. However, existing symbolic approaches introduce their own structural problems. Decision lists resolve conflicts through rule ordering, meaning that reordering the same rules can change the prediction [11]. Unordered rule sets avoid this but require external tie-breaking mechanisms [5](such as scoring, voting, or hand-tuned priorities) that lie outside the rules themselves and therefore outside the explanation. Exception Closed Conjunctive Rule Sets (ECCRS) [10] address both problems by encoding the relationship between defaults and exceptions directly in the structure of the rules. When two opposite-label rules can both apply to the same instance, one must be strictly more specific than the other. Prediction then follows a Most Specific Wins (MSW) principle: the most specific applicable rules determine the label. This makes explanations short, stable under reordering, and based off a visible chain of defaults and overrides that reflects how a human would justify the decision. An ECCRS ruleset can be translated into a KLM-style conditional knowledge base [9], and the same classification queries can then be answered using standard defeasible entailment methods. In this paper we will use System W inference on the same dataset used by Madzime [10] and validate the predictions made. An alignment theorem [10] guarantees that Rational Closure agrees exactly with MSW predictions under three structural conditions on the ruleset. In summary the main contributions of this paper are:

- Translating ECCRS into KLM conditional belief bases.
- Provides an implementation of System W entailment on translated belief bases, and verify its agreement with MSW predictions.
- Produce two-sided explanation traces for each classification query, reporting the ECCRS default-exception chain and the formal defeasible entailment side by side, and evaluate whether the traces are consistent.

In Section 2 we describe ECCRS. In Section 3 we provide a defeasible framework from which the implementation is built. In Section 4 we describe the translation of ECCRS rulesets to System W. Section 5 we shows the resulting evaluation results. Section 6 concludes.

2 Background: ECCRS

Each rule in ECCRS is structured as an if-then statement, allowing human readability [10]. Predictions can be justified by referencing the specific if clause that triggered the outcome. For each instance, a set of encoded rules indicates which rules were applied and which were overridden. This structure enables a precise justification to accompany each prediction.

ECCRS assumes a booleanised feature set of literals that are given based of a specific instance. ECCRS rule is a pair (B, l) where

- B is the conjunction of signed literals (the rule body)
- l is the class label either $\{0,1\}$

The rule (B, l) is only applicable if an instance satisfies all possible literals of B. The key structural relation of ECCRS is that if we have rule sets B and C then we have the following possibilities:

*

- B and C are said to be compatible if there are no such contradictory literals
- if B ⊂ C then C has more literals and is said to be more specific
- B and C are comparable if B ⊂ C or C ⊂ B.

Rulesets obtained by ECCRS are constrained such that any opposite labeled rules cannot conflict in any unstructured way. When two rules are applicable in the same instance and neither is more specific then a tie break is needed.

Definition 1:*Relaxed (local) closure.* A ruleset R is said to be relaxed exception closed in D if opposite labeled rules are required to be structurally comparable - when they overlap in D. Whenever a default and an exception can be imposed on the same row then the exception must be visibly more specific. Relaxed closure does not force them into co applying.

Definition 2:*Strict (global) closure.* R is said to be strictly exception closed if it is made comparable on the entire booleanised feature space. For all (B,*l*) and (C,1 - *l*) in R, their rule bodies must be comparable by strict inclusion. B ⊂ C or C ⊂ B. Strict global closure prohibits any opposite label conflicts in the feature space. The idea is that ECCRS will only be compatible with KLM translation if their are no conflicts. The RC reasoner is only defined of complete feature patterns.

Next we describe the Most Specific Wins operator.

Definition 3: Let F be an instance pattern such that $A(F) := \{$ (B,l) ∈ R: F ⊨ B $\}$ is the applicable set of rules.

Definition 4: The inclusion-maximal set of applicable rules is defined as: $M(F) := \{$ (B,l) ∈ $A(F)$: (C,l') ∉ $A(F)$, B ⊂ C$\}$

Most specific Wins is therefore defined as follows with respect to instance pattern F.

- MSW_R(F) = l if $M(F) \neq \emptyset$, ∀ M(F) have label l.
- MSW_R(F) = ⊥ if $M(F) = \emptyset$
- MSW_R(F) = ⊥ if $M(F)$ contains both labels

Example: Given a set of atoms p_1 .. p_5. We can a booleanised feature set as follow:

$$r_1 : (p_1) \Rightarrow 1$$
$$r_2 : (p_1 \wedge p_4) \Rightarrow 0$$
$$r_3 : (p_1 \wedge p_4 \wedge p_5) \Rightarrow 1$$
$$r_4 : (p_2 \wedge p_3) \Rightarrow 1$$
$$r_5 : (p_2 \wedge p_3 \wedge p_4) \Rightarrow 0$$

If we define a feature pattern F := $p_1 \wedge \neg\, p_2 \wedge \neg\, p_3 \wedge p_4 \wedge p_5$. We see that we have an exception to an exception such that M(F) = $\{r_3\}$ and MSW_R = 1

- r_1 applies
- r_2 overrides r_1
- r_3 overrides r_2

Example 2: We define a feature pattern F := $p_1 \wedge \neg\, p_2 \wedge \neg\, p_3 \wedge p_4 \wedge \neg\, p_5$. Since p_5 is false we see that r_2 is the most specific that applies thus MSW_R = 0.

Example 3: We define a feature pattern F := $\neg\, p_1 \wedge p_2 \wedge p_3 \wedge \neg\, p_4 \wedge \neg\, p_5$. The only applicable rule is r_4. Thus MSW_R = 1.

Example 4: If no rule pattern is uncovered it must be the case that no rule body is satisfied. Thus MSW_R (F)= ⊥ .

Considering these four outcomes we can safely conclude that given atoms of any kind there is sufficient justification for the prediction. The most specific wins principle allows one to see exactly which rule justified the conclusion.

Furthermore, we now show the three assumptions required by an ECCRS ruleset so that it can be translated into a KLM knowledge base $K(R)$. These assumptions are as follows:

1) Strict global exception closure: For any compatible opposite-label rules (B, l) and (C, 1 − l) in R, we require: $B \subset C$ or $C \subset B$

2) Sanity: requires that two rules may not have the same body and opposite labels.

3) No total override: requires that every rule must have at least one completion of its body, where it is not less specific that a rule with the opposite label. For every (B, l) ∈ R there must exist a complete feature pattern F such that:

F ⊨ B and ∀ (C, 1 − l) ∈ R , ($B \subset C \implies F \not\models C$).

Alignment theorem

Let $K := K(R)$ be the translated KB. Then for every complete feature pattern F:

(1) If no ECCRS rule applies to F, then SW entails *neither* $F \mid\sim m$ nor $F \mid\sim \neg m$.
(2) If $\mathrm{MSW}_W(F) = 1$, then SW entails $F \mid\sim m$ and *does not* entail $F \mid\sim \neg m$.
(3) If $\mathrm{MSW}_W(F) = 0$, then SW entails $F \mid\sim \neg m$ and *does not* entail $F \mid\sim m$.

ECCRS	SW query	Output
1	$K(R) \mid\approx_{\mathrm{W}} F \mid\sim m$	class 1
0	$K(R) \mid\approx_{\mathrm{W}} F \mid\sim \neg m$	class 0
⊥	neither	undecided

3 Defeasible Reasoning and System W

Propositional Logic is the basic language chosen by Kraus, Lehmann and Magidor when defining the KLM framework [9]. First, we must introduce the relevant syntax of this logic before we can express the ideas related to defeasible reasoning. A propositional atom is a statement that can be assigned one of two values, conventionally called true and false or T and F.[2] Statements are represented by a meta variable: $p, q, r, \ldots$ such that b could mean the statement that "birds fly". The set of *atomic propositions* is denoted by P. Formulas are created by combining atoms with the set of connectives { ∧, ∨, ¬, →, ↔ }. All connectives are binary except for the unary ¬. Each connective will accept all formulas as arguments. With this in mind we can now explore more expressive formulas. The set of all formulas is denoted as $\mathcal{L}$, which has elements {α,β,γ, ...}. A formula in propositional logic is defined recursively[2] such that for any atom in P and any α,β in $\mathcal{L}$ we can formulate: $\alpha : p, \neg\, \alpha$, $\alpha \wedge \beta$, $\alpha \vee \beta$, $\alpha \rightarrow \beta$, $\alpha \leftrightarrow \beta$. α is said to be the antecedent and β the consequent. The two constants defined in propositional logic follow that of tautology and contradiction. Denoted by ⊤ and ⊥ , respectively and being elements of the language $\mathcal{L}$. An interpretation for A is a total function $P_A : P_A \mapsto \{$ T, F $\}$ that assigns one of the truth values T or F to every atom in P_A. This is known as a valuation but is referred to as worlds or interpretations in the literature [7]. If an atom is true in a valuation, then it is said that the valuation satisfies the atom. This is denoted by ⊩ such that for

some valuation u, if it is the case that for p ∈ P, $u(p) = T$, then u ⊩ p. Similarly if $u(p) = F$, then u ⊮ p. This can be extended for all α, β, γ,... ∈ $\mathcal{L}$. We define $\mathcal{U}$ to be the set of all valuations in $\mathcal{L}$. Given the set of valuations we can define further that for any $\alpha \in \mathcal{L}$, let $\hat{\alpha}$:= { $u \in \mathcal{U} \mid u \Vdash \alpha$ }. For any $u \in \hat{\alpha}$, u is referred to as a model of α. A formula α is considered satisfiable if there exists a $u \in \mathcal{U}$ such that $u \Vdash \alpha$. A formula α is considered unsatisfiable if there is no such u $\in \mathcal{U}$ such that $u \Vdash \alpha$.

The limitations regarding propositional logic are mainly focused on exceptions within knowledge bases. One way this can be remedied is by denouncing monotonicity as a property. There are many defeasible approaches in the literature and they are all non-monotonic. The preferential approach to non-monotonic reasoning is central to that of the KLM framework. It was first defined by Shoham [13] and later extended by Kraus et al [9]. The notion of entailment using preferential semantics can be defined by the statement that $\alpha \mid\sim \beta$ which reads as "α preferentially entails β", iff all the preferred models of α also satisfy β. Another way to describe this relationship is that α "typically" holds given β. Ranked Interpretations can be thought of as multiple tiers of possible states containing any number of interpretations. Ranked interpretations is a subset of preferential ordering. The only difference being that ranked interpretations do not hold duplicate states for specified valuations. If any two states share a valuation then one of the states will be irrelevant. A ranked interpretation can be thought of as a function with a set of valuations as input and a totally ordered set as output (minimal valuations are considered only). We will not discuss the semantics for rational closure but rather focus on the algorithm for rational closure (RC). Firstly, one must define the BaseRank algorithm. RC is the most conservative of the defeasible reasoning algorithms. It is of the prototypical variety which means it only takes the minimal set of inferences given a Defeasible Knowledge Base, $\mathcal{K}$. Here is the algorithm of BaseRank outlined by Casini et al. [4]

Algorithm 1 BaseRank

Input: *A knowledge base* $\mathcal{K}$
Output: *An ordered tuple* $(R_0, ..., R_{n-1}, R_\infty, n)$

$i := 0$;
$E_0 := \overrightarrow{\mathcal{K}}$;
repeat
  $E_{i+1} := \{\alpha \rightarrow \beta \in E_i \mid E_i \models \neg\alpha\}$;
  $R_i := E_i \setminus E_i + 1$;
  $i := i + 1$;
until $E_{i-1} = E_i$;
$R_\infty := E_{i-1}$;
if $E_{i-1} = \emptyset$ **then**
  $n := i - 1$;
else
  $n := i$;
end if
return $(R_0, ..., R_{n-1}, R_\infty, n)$

The above algorithm assigns a ranking to each statement in K based on their exceptionality. Which is any α that makes the knowledge base unsatisfiable. The formula α is said to be exceptional in the set K iff $\{\alpha_1, ... , \alpha_n\} \cup \{\neg\alpha\}$ is unsatisfiable. $\overrightarrow{\mathcal{K}}$ is the materialisation of $\mathcal{K}$ defined as $\overrightarrow{\mathcal{K}} := \{\alpha \rightarrow \beta \mid \alpha \mid\sim \beta \in \mathcal{K}\}$. Classical statements are given the highest ranking and can never be excluded. Defeasible statement are ranked from most broad to least. Exceptional statements are given the higher ranks as they may violate rules that are more broad. The second part of rational closure is to determine whether a knowledge base entails a defeasible implication. Given an input of a defeasible implication we check it for any conflicts against each rank created from the knowledge base. If we find a rank that does not conflict with the statement than we can predict it typically holds and in what world it does. If the defeasible implication always conflicts we cannot deduce that it entails the knowledge base. The algorithm is defined in Algorithm 2:

Algorithm 2 Rational Closure

Input: *A knowledge base* $\mathcal{K}$ and a defeasible implication $\alpha \mid\sim \beta$
Output: **true** if $\mathcal{K} = \mid\approx \alpha \mid\sim \beta$ and **false** otherwise

$(R_0, ..., R_{n-1}, R_\infty, n)$:= BaseRank($\mathcal{K}$);
$i := 0$;
$R := \bigcup_{j=0}^{j<n} R_j$;
while $R_\infty \cup R \models \neg\alpha$ **and** $R \neq \emptyset$ **do**
  $R := R \setminus R_i$;
  $i := i + 1$;
end while
return $R_\infty \cup R \models \alpha \rightarrow \beta$;

In the case of System W we can define that a propositional signature is a finite set Σ of propositional variables. The language over Σ is denoted by $\mathcal{L}_\Sigma$. We denote elements of a signatures with lowercase letters a,b,c, ... and formulas with uppercase letters A,B,C, ... An interpretation over a signature is denoted as Ω_Σ. An interpretation $\omega \in \Omega_\Sigma$ is a model of a formula A $\in \mathcal{L}_\Sigma$ if A holds in ω. A set of modesl of a formula over a signature is denoted as $Mod_\Sigma(A) = \{\omega \in \Omega_\Sigma | \omega \models A\}$ or as Ω_A. A formula A entails a formula B if $Mod_\Sigma(A) \subseteq Mod_\Sigma(B)$. [1] A conditional (B|A) corresponds to "given A then usually B holds." . Thus A is the antecedent and B the consequent. A conditional belief base is a finite set of conditionals denoted by Δ . It is consistent if it has a model. Reasoning using conditionals is modelled by inference relations. An inference relation is a binary relation $\mid\sim$ on formulas over a signature Σ with A $\mid\sim$ B means that A "typically" entails B. An an inductive operator such as System W is said to map a belief base to an inference relation [8]. The algorithm for System W was modelled with Partial MaxSAT by Beierle et al.[1] This was the algorithm used within this paper to compare alignment with the learnt ECCRS rulesets. The algorithm for System W inference takes a belief base Δ and two formulas A,B as input. In our case, the formulas would include the ECCRS ruleset of conditionals as formula A and formula B would be the instance query with their feature set and given prediction label y. The System W inference will compare whether the ruleset entails the same prediction as MSW. The System W inference will run two queries of the possible labels for y (i.e , y and !y). In either case the System W inference will have four possible outcomes:

- only y entailed: the reasoner infers label 1
- only not y entailed: the reasoner infers label 0

- neither: the reasoner is undecided
- both: the instance is inconsistent with the knowledge base, i.e the assumptions have been violated.

To illustrate the algorithm used for the System W inference implemented by Beierle et al [1] we much introduce some more background. Firstly we define and ordered parition $OP(\Delta)$ or tolerance partition[6]. Which means that for any conditional there exist a world $\omega \in \Omega_\Sigma$ such that ω does not falsify any conditional in Δ and verifies a conditional. An ordered partition of a belief base is the partition of Δ where each Δ_i is the maximal subset of the $(\Delta_1 \cup \Delta_2 \cup ... \cup \Delta_k)$ and is tolerated by $(\Delta_1 \cup \Delta_2 \cup ... \cup \Delta_k)$ [1].
For a formula F and a set of formulas M over Σ:
$\omega \models M$ iff $\omega \models F_i$ for every $F_i \in M$
$\Omega_M = \{\omega \in \Omega_\Sigma \mid \omega \models M\}$
$\overline{M} = \{\overline{F_i} \mid F_i \in M\}$
$F \wedge M = F \wedge F_1 \wedge ... \wedge F_M$ for $M = \{F_1, ..., F_m\}$
For a conditional (B|A), the formual $\overline{A} \wedge B$ expressing its non-falsification is denoted by $nf(B|A)$, and nf is extended to a set of conditionals Δ. Thus,

$$nf(\Delta) = \{\overline{A} \vee B \mid (B|A) \in \Delta\}$$
$$\overline{nf(\Delta)} = \{A \wedge \overline{B} \mid (B|A) \in \Delta\}$$

are the sets of falsifying and non-falsifying formulas, for the conditionals in the belief base. The SWinf algorithm requires that we need two sets of formulas that act as hard and soft constraints. The soft constraints are all formulas represented in S and the hard constraints are all formulas represented in H. The MSS is called the maximal satisfiable subset, and the MCS is the minimal correction subset. A set N is called a minimal correction subset (MCS) w.r.t to (S,H). if $S \setminus N$ is a MSS w.r.t (S,H).
A MSS is a subset of S such that there exist an ω in H that entails M for every $M' \subseteq S$ and $M \nsubseteq M'$ that could satisfy a given set M w.r.t (S,H). Let us consider, now the algorithm for System W inference implemented by Beierle et al [1].
The inference algorithm of System W works by comparing sets through inclusion whereby it removes sets that are no longer tolerated. The main part of the algorithm is the recursive function recWinf which takes an index j pointing to a given conditional Δ^j in the ordered partition $OP(\Delta)$ and $nf/f - condition$ for (Δ, j) as input. V is filled with non-falsifying formulas for the conditional in the partition element as soft constraints, and the hard constraints is the verification formula of $A \wedge B$ which will be enlarged by H_{new} for every subsequent call. The main recursive loop checks if $V \cup F$ is empty if so we can return true otherwise we must check partition elements that are lower down in $OP(\Delta)$. If we are already at the lowest partition element then $recWinf$ and also SWinf will return false. Otherwise a new $nf/f - condition$ for $(\Delta, j-1)$ is created. All recursive calls must return true for A $\mid\!\approx_{\mathrm{W}}$ B.

Algorithm 3 SWinf(Δ, A, B)

Input: A belief base Δ and formulas A, B
Output: **true** if $A \mid\!\approx_W B$ and, **false** otherwise

```
1:  OP(Δ) = (Δ_1, Δ_2, ..., Δ_k)
2:  function recWinf(j, H)(1)
3:      V := MCS(nf(Δ^j), H ∪ {A ∧ B});
4:      F := MCS(nf(Δ^j), H ∪ {A ∧ B̄});
5:      if ¬(∀N' ∈ F∃N ∈ V.N ⊆ N' then return false
6:      end if
7:      for N ∈ V ∩ F do
8:          if j = 0 then return false
9:          end if
10:         H_new := (nf(Δ^j)\N) ∪ N̄;
11:         if recWinf(j − 1, H ∪ H_new) = False then return false
12:         end if
13:     end for
14: return true
15: end function
16: return recWinf(k, ∅);
```

4 ECCRS implementation and translation for System W inference

This section looks at how ECCRS is specifically computed and predictions made using the MSW operator. Furthermore, this section will show how the ECCRS rulesets are further translated to conditional belief bases that can be used by System W to map the belief bases to an inference relation or prediction. The ECCRS works on a dataset where every feature of an instance is described by a binary input (1 or 0) and a binary label final result label y. We can compute these against the learnt ECCRS ruleset to find the generated rules that reflect why the chosen label appears. We are able to see the exception-default chain of MSW, showing which rules were applied to create the final prediction. The alignment result will rely on 3 assumptions about an exported ruleset R for its translation into a KLM knowledge base K(R) [10]. We define one atom for the whole knowledge base K. Let m be a singular fresh propositional atom. It represents the positive class of entailments. Thus, m is equivalent to the label 1 and not m means label 0. This will turn every ECCRS rule into a defeasible conditional: a rule (B,1) becomes B $\mid\sim$ m and a rule (B, 0) becomes B $\mid\sim$ not m meaning label 0. From **Example**: above the rules will be translated into:

$$r_1 : p_1 \mid\sim m$$
$$r_2 : p_1 \wedge p_4 \mid\sim \neg m$$
$$r_3 : p_1 \wedge p_4 \wedge p_5 \mid\sim m$$
$$r_4 : p_2 \wedge p_3 \mid\sim m$$
$$r_5 : p_2 \wedge p_3 \wedge p_4 \mid\sim \neg m$$

Using the defeasible version of the rules from ECCRS we can use them and rank each of them using the BaseRank algorithm. The materialisation is defined by represented classical entailment of the set of formulas representing our rules. The materialisation is as follows:

$$r_1 : p_1 \rightarrow m$$
$$r_2 : p_1 \wedge p_4 \rightarrow \neg m$$
$$r_3 : p_1 \wedge p_4 \wedge p_5 \rightarrow m$$
$$r_4 : p_2 \wedge p_3 \rightarrow m$$
$$r_5 : p_2 \wedge p_3 \wedge p_4 \rightarrow \neg m$$

The reason we want the materialisation to be the classical implication is so that we can conduct entailment check on the conditional directly using a satisfiability solver. In our case we are using Z3 as stated in the infOCF library [1, 3]. Algorithm 1 is used to create the ordered partition in the form of conditionals. The conditionals created by and instance pattern will take the form $(m|F)$ where F is the instance pattern provided by a single record of the dataset and m represents the prediction label. The datasets used are that publicly available from the UCI Machine Learning Repository [14]. The Breast Cancer Wisconsin dataset was used which predicts whether a tumor is malignant or benign (175 instances and, 9 features). Mushroom (2031 instances, 22 features), and Bank Marketing (11303 instances, 16 features). One-hot encoding was used afterwhich each data point is represented as a numeric 1 or 0. We first compute the ECCRS MSW directly on each of the datasets. Every instance or record in the data will correspond to values for each of the a_i boolean features. Using the learnt rules we can compute each overridden rule and what rule made MSW come to its final prediction whether it agrees with that instance or fails to make the correct prediction. We will gather the applicable rules first matching each feature to its corresponding rule and then choose the most specific of the applicable rules this is done algorithmically as follows: Given that we have found all the applicable rules, we must simply

Algorithm 4 applicableRules

Input: An instance list I, A rule list R
Output: List of applicable rules A

```
A := [];
for instance in I do:
    tmp = []
    for rule in R do
        if isSubset(rule.cond, instance.cond) then
            tmp := [tmp, rule.id]
        end if
    end for
    A.append(tmp,instance.id)
end for
return A
```

compute the MSW of the given applicable rules. The ECCRS MSW prediction will use the label that requires the greatest number of binary features. The assumption validation follows whereby strict global exception closure implies sanity [10] and no total override is checked by inspecting rule bodies and labels. A rule (B,l) is totally overridden in a rule set S if every completion of B satisfies some stricter opposite-label rule in S. The datasets used in this paper do not violate any assumptions [10]. The translation to a conditional belief base is defined and implemented as follows:
A signature Σ is created by combining the label with all features of all rulesets R created from a each dataset. Thus, each dataset will have its own signature. Each rule will be represented as its own conditional in the form: $(m|F)$, m is true given that the instance F is true. F represents all the $a_i's$ in the feature. Using the signature Σ and each instance as a query we can run System W inference on both possible label values. The output of which will follow:

Dataset	Entailed	Not Entailed	Abstain	Violation
Cancer	107	68	0	0
Mushroom	977	1054	0	0
Bank	331	10972	0	0

Table 1: Evaluation of 4 outcomes from System W inference using ECCRS Rulesets and both possible prediction labels

Dataset	MSW:1	MSW:0	Dataset:1	Dataset:0	Accuracy
Cancer	107	68	115	60	0.942
Mushroom	977	1054	979	1052	0.999
Bank	331	10972	1322	9981	0.897

Table 2: Evaluation of ECCRS computation on instances from all 3 datasets. Accuracy shows MSW prediction with the true values from the dataset

ECCRS	SW query	Output
1	$\Delta(R) \mid\!\approx_W (m\|F)$	class 1
0	$\Delta(R) \mid\!\approx_W (!m\|F)$	class 0
$\perp$	neither	undecided

The signature size for each dataset varies and the size of queries are equal to the number of instances multiplied by two for each dataset. The Breast Cancer Wisconsin Dataset has 7 conditionals and 350 queries, Mushroom had 11 conditionals and 4062 queries and Bank had 15 conditionals and 22606. The number of checks will grow at an exponential rate relative to the size of the signature. The evaluation was executed on an AMD Ryzen 5-3600 CPU with 32GB DDR4-3200 working memory.

5 Evaluation Results

Table 1 summarizes the final count for each query inference of System W and each case resulted in either a rejection of entailment or entailment. None of the datasets produced cases whereby the inductive inference operator abstained or produced results relating to an inconsistent belief base. This is to be expected as the datasets used satisfied the three assumptions mentioned. Table 2 summarises the ECCRS MSW computation and its predictions given the same datasets. From Table 1 and 2 it can be seen that agreement rate for all datasets between the MSW prediction and the System W prediction is 1. It follows that if a ruleset R satisfies strict global exception closure, sanity and no total override then for every complete F, System W entails the same label as $MSW_R(F)$ [10].

6 Conclusions

In this paper, we implemented both System W using infOCF library with a current MaxSAT solver (Z3) and we implemented ECCRS MSW side by side. From the results we can conclude that there is sufficient evidence that MSW predictions are supported by System W given that a dataset satisfies the three assumptions. In summary, we have translated ECCRS rulesets into conditional belief bases that can be used as System W inference and we have shown that given the assumptions are satisfied, System W provides consistent results to MSW predictions. In conclusion, the systematic approach

in defining the datasets and providing complete instance patterns allows us to guarantee that a symbolic classifier like ECCRS can be proven to be logically rigorous and thus provide explanations that directly correspond to specific logical data points. Defeasible machine learning hopes to have interpretable classifiers that show users which rules fire. Thus, explanations are presented in the structure of the predictions rather than projected onto the predictions.

References

[1] Christoph Beierle, Aron Spang, and Jonas Haldimann. 2024. A Partial MaxSAT Approach to Nonmonotonic Reasoning with System W. *The International FLAIRS Conference Proceedings* 37, 1 (May 2024). doi:10.32473/flairs.37.1.135330

[2] Mordechai Ben-Ari. 2012. *Mathematical Logic for Computer Science.* Springer London. https://doi.org/10.1007/978-1-4471-4129-7

[3] Nikolaj Bjørner, Leonardo de Moura, Lev Nachmanson, and Christoph M. Wintersteiger. 2018. Programming Z3. In *Engineering Trustworthy Software Systems: 4th International School, SETSS 2018, Chongqing, China, April 7–12, 2018, Tutorial Lectures* (Chongqing, China). Springer-Verlag, Berlin, Heidelberg, 148–201. doi:10.1007/978-3-030-17601-3_4

[4] Giovanni Casini, Thomas Meyer, and Ivan Varzinczak. 2019. *Taking Defeasible Entailment Beyond Rational Closure.* 182–197. doi:10.1007/978-3-030-19570-0_12

[5] Adnan Darwiche and Auguste Hirth. 2023. On the (complete) reasons behind decisions. *Journal of Logic, Language and Information* 32, 1 (2023), 63–88.

[6] Moisés Goldszmidt and Judea Pearl. 1996. Qualitative probabilities for default reasoning, belief revision, and causal modeling. *Artificial Intelligence* 84, 1 (1996), 57–112. doi:10.1016/0004-3702(95)00090-9

[7] Adam Kaliski. 2020_. An overview of KLM-style defeasible entailment. http://hdl.handle.net/11427/32743

[8] Christian Komo and Christoph Beirle. 2022. Nonmonotonic reasoning from conditional knowledge bases with system W. *Annals of Mathematics and Artificial Intelligence* 90, 1 (2022), 1–60. doi:10.1007/s10472-021-09777-9

[9] Sarit Kraus, Daniel Lehmann, and Menachem Magidor. 1990. Nonmonotonic reasoning, preferential models and cumulative logics. *Artificial Intelligence* 44, 1 (1990), 167–207. doi:10.1016/0004-3702(90)90101-5

[10] Ruvarashe Madzime and Thomas Meyer. 2026. Toward Defeasible Machine Learning: When Learned Rule Sets Become Defeasible Theories. (2026). https://ceur-ws.org/Vol-4239/paper-Madzime-3.pdf.

[11] Fernando E. B. Otero and Alex A. Freitas. 2013. Improving the Interpretability of Classification Rules Discovered by an Ant Colony Algorithm. In *Proceedings of the Fifteenth Annual Conference on Genetic and Evolutionary Computation (GECCO '13).* ACM, 73–80.

[12] Cynthia Rudin. 2019. Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead. arXiv:1811.10154 [stat.ML] https://arxiv.org/abs/1811.10154

[13] Yoav Shoham. 1987. A Semantical Approach to Nonmonotonic Logics. In *Proceedings of the Second Annual IEEE Symposium on Logic in Computer Science (LICS 1987)* (Ithaca, NY, USA). IEEE Computer Society Press, 275–279.

[14] WIlliam Wolberg. 1990. Breast Cancer Wisconsin (Original). UCI Machine Learning Repository. DOI: https://doi.org/10.24432/C5HP4Z.